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AI-Driven Data Engineering Project

Team No: 1
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Project Title:

AI-Driven Log Anomaly Detection Using Hadoop and Apache Spark

Abstract:

Modern distributed computing environments generate massive volumes of system log data that record events such as system operations, errors, and performance metrics. Analyzing these logs manually becomes difficult as the scale of data grows, making it challenging to detect abnormal system behavior in a timely manner. This project presents an AI-driven log anomaly detection system using Hadoop and Apache Spark to address this challenge.

The proposed system stores large-scale log datasets in the Hadoop Distributed File System (HDFS) and performs distributed log processing and feature extraction using Apache Spark. To automatically identify unusual patterns in the logs, the Isolation Forest algorithm, an unsupervised machine learning technique, is applied to detect anomalies without requiring labeled data. The system analyzes log characteristics such as error frequency, response time, and event patterns to identify deviations from normal behavior.

The results provide insights into system health, including anomaly percentages and error distributions. By integrating artificial intelligence with scalable big data technologies, the proposed approach offers an efficient and automated solution for monitoring large-scale systems and detecting potential failures at an early stage.

Vrinda
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